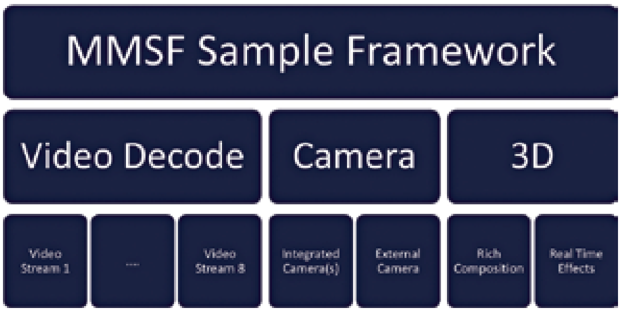
of multiple power and thermal technologies. The high-level architecture of Intel DPTF is shown in Figure 1.

Intel DPTF is a BIOS ACPI implementation on specific devices/systems. Technologies contained within the framework are mainly focused on thermal management per the ACPI 5.0 specification. The thermal management technology contained in Intel DPTF is called Intel® Dynamic Power Performance Management (Intel® DPPM). Intel DPPM is an intelligent closed-loop hardware and software solution designed to manage platform thermals in a constrained design. An OEM design team controls platform skin and component temperatures, allowing them to design for a "typical" rather than "worst case" power scenario and minimize any performance impact. Intel DPPM technology has a series of policies that can be used to more efficiently control platform behavior when a thermal event is triggered, see Table 1.

GAME PERFORMANCE IMPACT IN TABLET MODE ON 2 IN 1 DEVICES

A very small number of game developers have reported performance issues with 2 in 1 devices in tablet mode that are attributed to Intel DPTF. In tablet mode Intel DPTF will often use a passive policy, meaning available power is reduced so the device stays at a comfortable thermal level for handling. Whereas in laptop mode the device is often sitting on a surface and skin temperature is less of a concern. The processor increases frequency in response to application need by utilizing Intel® Turbo Boost Technology. When increasing above the base frequency there are 2 limits, the max frequency (defined by the OEM) and the power available. As Intel DPTF has lowered the available power to keep the device from getting too warm to comfortably hold, it also restricts how far Intel® Turbo Boost Technology can increase the frequency and thus lower game performance.

Intel DPTF is a power and thermal management solution designed to extend battery life for mobile devices. Often the key to extended life is lower power usage, limited power means lower operating frequencies. Something that is not an issue with common applications, but in rare cases may have an impact on game performance in tablet mode. Game performance is not always impacted as a result of Intel DPTF, but it is possible. If you are observing a performance difference between tablet and laptop modes on a 2 in 1 device with your game, there is no function within the graphics driver or operating system to disable Intel DPTF. If the OEM's design team included the option to disable Intel DPTF, consumers can disable it, but doing so will sacrifice battery life. Disabling Intel DPTF requires the consumer to enter the configuration section of their device's BIOS.



Multi-Media Sample Framework Architectural Components

MULTI-MEDIA SAMPLE FRAMEWORK (MMSF)

The Multi-Media Sample Framework provides a robust set of basic functionality that can help you develop rich media applications. It accomplishes this by delivering a simple unified environment from which to access and explore video decode, 3d graphics, real-time frame/image processing, and camera capabilities available on Intel® processor-based platforms, either Windows or Android. Developers working in the areas of local video playback, streaming video playback, multi-stream video playback, image processing, video processing, or camera-centric applications in general may find the Android and/or Windows MMSF samples useful. In particular the MMSF focuses on offloading as much work as possible to either fixed function hardware or the graphics processing unit. This results in very low CPU utilization at runtime for both single and multi-stream video playback cases and opens the door for developers to leverage additional processing on the CPU if needed.

MMSF makes use of three core capabilities present on most modern operating systems. These include: video decode, camera, and 3d graphics. Please note that depending on the operating system, the libraries which expose the capabilities described will be different.

FEATURES

- Supported Operating Systems - Android* 4.4.x and higher, Windows* Vista*, 7, 8 and higher.
- Video Decode - Provides a basic video decode pipeline. Single or multiple streams may be decoded concurrently.
- Camera Feed - Provides integrated camera capabilities that allow capture, processing, and display of camera data.
- Pre-Processing - Provides a simple pixel/fragment shader based mechanism for processing and manipulating camera data before it is displayed.
- Post-Processing - Provides a simple pixel/fragment shader based mechanism for processing decoded video frame data before it is displayed.
- Display to 3D Geometry - Provides capabilities to display video information on scalable 3D geometry.

INTERNET OF THINGS

Internet is the next level of evolution towards developing connected devices. With the help of unique hardware and developer kits along with sensors available from Intel you can create smart devices that interconnect with each other and automate your life like you never imagined before. The developer kit for IoT includes a development board and starter kit including Yocto Linux system, Eclipse and XDK IDE, IoT Cloud Analytics, a set of libraries, and more. Developers can choose to order either the Intel® Galileo or Intel® Edison board and the Grove Starter Kit Plus - Intel® IoT Edition. More details like the getting started guide, documentation, project gallery and forums are available at http://maker.intel.com.

WHAT'S INSIDE THE GROVE STARTER KIT PLUS - INTEL® IOT EDITION:

Base Shield v2
Grove - Buzzer

The Grove Starter Kit Plus - Intel® IoT Edition